ZapBox Headless – Operating Instructions

Language: English | **Version:** oi940432

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Overview

The **ZapBox Headless** is an electronic switch for Bitcoin Lightning payments without a display. A payment via the Lightning Network can switch an output – ideal for embedded applications, concealed installations, mechanical engineering, and anywhere a display is not required.

The operating state is indicated exclusively via a **status LED**. A second **action LED** indicates the switching function.

Basic Equipment

Component	Description
Microcontroller	ESP32 Dev Module (no display)
Input	USB-C (Power IN)
Output	USB-C (Power OUT)
Status indicator	Status LED with blink patterns and action LED as feedback
Control	Micro button for BOOT (config mode) and reset
Switch	3-position slide switch (AUTO / OFF / ON)
Option NFC Module	For Bolt Cards (NTAG424 DNA) and standard NTAG213/215/216 (with LNURL-withdraw)

Views

Image 1: Front view / top view

Connections

Input – USB-C Socket for Power Supply (5V)

Power the device via the **Power IN** connector with a USB-C cable at **5 V DC (max. 5 A)**.

> **Note:** The Power IN connector is not "intelligent". Some chargers or power modules with a USB-C output do not recognize the ZapBox and will not supply power. In this case, use a **USB-A output** of the power supply or a different power source.

Input – Micro-USB Socket on the Microcontroller (Data Access, front)

To read data from or transfer data to the device, connect the ZapBox to a computer or laptop:

1. On the **front left**, there is a small panel to the left of the USB connectors. Open the panel by sliding it to the right from below using a **narrow screwdriver**.
2. Connect a Micro-USB cable to the microcontroller.

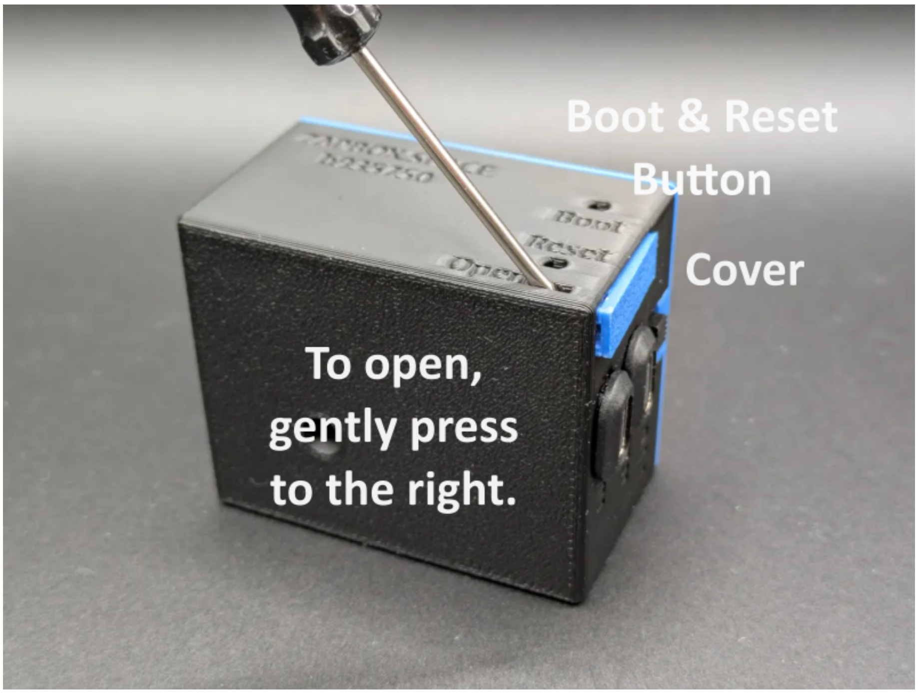

Image 2: Opening the panel for data connection

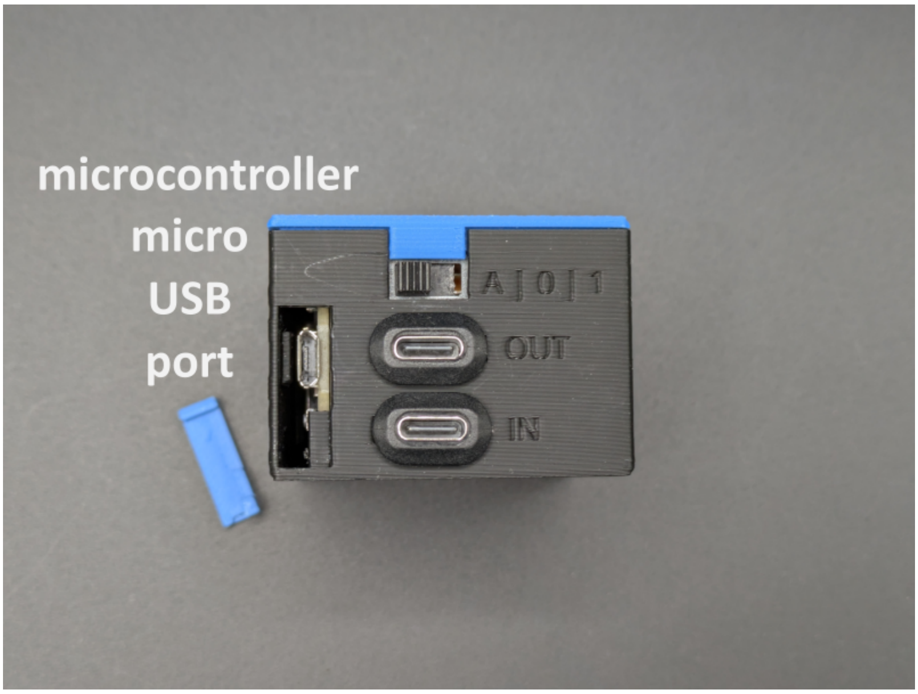

Image 3: Micro USB port

> **Important Note:** The USB connector directly on the microcontroller is intended exclusively for flashing the firmware and transferring configuration parameters. During the flashing process, no load may be connected or switched at the output, as this can cause malfunctions or **damage to the microcontroller**.
>
> It is therefore recommended:
>
> - not to connect any load to the output during the USB connection, or
> - to additionally connect the regular **Power IN input** to the same power supply. This ensures that the current for the power relay does not flow through the microcontroller and overload it.

Output – USB-C Socket (Switched 5V Voltage)

The USB socket is switched via a relay switching contact. The **total load** of the sockets should **not exceed 3 A**.

Controls

The ZapBox Headless has **no display**. The operating state is indicated exclusively via the **status LED** and **action LED**.

Status LED – Blink Patterns

Pattern	Meaning
3× short blinks at startup	Boot completed
Rapid blinking	Establishing connection / initialization
Slow blinking (1 Hz)	Config mode active
Steady light	Ready for operation, waiting for payment
200 ms on / 800 ms off	NFC payment pending
2× short blinks	Payment successful
3× short blinks	NFC timeout / error
1× blink (500 ms on/off, 2 s pause)	Error pattern 1: No Wi-Fi
2× blinks (300 ms on/off, 2 s pause)	Error pattern 2: No internet
3× blinks (250 ms on/off, 2 s pause)	Error pattern 3: Server unreachable
4× blinks (200 ms on/off, 2 s pause)	Error pattern 4: WebSocket connection failed

Control Buttons

Function	Button
Enter config mode	Hold BOOT button for at least 5 seconds
Restart	Reset button

Triple Slide Switch (AUTO / OFF / ON)

Position	Function
A (AUTO)	Automatic mode – Normal operation
0 (OFF)	Power supply interrupted – Output OFF
1 (ON)	Output (USB-C) permanently ON

Setup and Commissioning

The ZapBox is tested after manufacturing and delivered with the current firmware – however, it is not yet configured. The software is actively developed, so it is recommended to flash the ZapBox with the latest firmware right from the start and then perform a configuration. There is a convenient **Web Installer** (Headless version) for this purpose.

Here is a step-by-step guide for setup:

1. Connect the ZapBox to the Micro-USB port of the microcontroller with a cable and connect it to a computer.
2. Open a Chromium browser, for example Google Chrome, Microsoft Edge, Brave, Vivaldi, Opera, or Helium.
3. Navigate to the Web Installer page **https://installer.zapbox.space/headless/**.
4. Flash the latest "Latest" version (Headless), as described in step 1 of the Web Installer.
5. After flashing, close the small window and go to step 3 – Load config values. There, click the `Connect` button.
6. You should now see `Connected` and `Config mode` in the green field. Config mode is active as soon as the **status LED blinks slowly** (approx. 1 Hz). If not, check step 2 – Prepare connection.
7. The ZapBox requires three parameters: `WiFi SSID` / `WiFi password` / `Device settings string`. You get the device settings string from your LNbits wallet. Add the **Bitcoin Switch** or **ZapBox** extension for this. The ZapBox extension also supports the NFC module; otherwise they are identical.
8. After all three parameters have been entered, save them using the `Write Config` button and restart the ZapBox with the `Restart` button.

After initialization, the status LED lights up steadily – the ZapBox is ready for operation and waiting for the first payment.

In case of errors or malfunctions, please refer to the "Error Detection & Report" and "Troubleshoot" chapters further down on the Web Installer page.

Option: NFC Module

Depending on the configuration, an NFC module may be installed on the **top of the ZapBox**. Alternatively, the module is also available separately. The ZapBox currently supports the following card types:

- **Boltcards** (NTAG424)
- **LNURL-Withdraw** from NTAG21x (213 / 215 / 216)

The LNbits ZapBox Extension is required for this function. It must be supported by the LNbits server.

Technical Specifications

Property	Value
Supply voltage	5 V DC via USB-C
Maximum input current	5.0 A
Output power	max. 3.0 A (recommended)

Property	Value
Microcontroller	ESP32 Dev Module (WROOM-32)
Flash memory	4 MB
SRAM	512 KB
Display	none (Headless)
Status indicator	Status LED / Action LED
Communication	Wi-Fi (ESP32)
Relay channels	1 – Expandable up to 12 (CH01–CH12)
Payment protocol	Bitcoin Lightning Network

Safety Information

- Operate the device exclusively with the specified supply voltage.
- Do not exceed the maximum current loads of the outputs.
- Do not perform any work on the relay contacts under load.
- The device is not suitable for use in humid or wet environments.
- Keep out of reach of children.

Further Links

Resource	Link
Overview of all ZapBox models	https://zapbox.space/
Web Installer, quick overview & troubleshooting	https://installer.zapbox.space/headless/
Detailed documentation (parameters & functions)	https://ereignishorizont.xyz/zapbox/
GitHub repository (software, PCB layouts, 3D print files, operating instructions)	https://github.com/AxelHamburch/ZapBox
ZapBox Extension	https://github.com/AxelHamburch/zapbox_extension
LNbits	https://lnbits.com/

Subject to changes and errors. As of: 2026